

ECEN-662: Homework 3

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Problem 1

Let $X = (N, Y)$, where N is a random variable taking values from $\mathbb{Z}_{++}$ with a fixed pmf $p(n)$. Having observed $N = n$, mother nature performs n Bernoulli trials with success probability θ , observing a total of Y successes.

- i) Show that X itself is a minimal sufficient statistic for θ .
- ii) Show that $\hat{\theta}(X) = \frac{Y}{N}$ is an unbiased estimator of θ and

$$\text{Var}_{\theta}[\hat{\theta}(X)] = \theta(1 - \theta)\mathbb{E}\left[\frac{1}{N}\right].$$

Hint: It may be useful to consider the law of total variance.

Solution

Part (i): Since the ratio of $p(x)$ and $p(y)$ is independent of θ when $x = y$, X is a minimal sufficient statistic for θ .

Part (ii): First

$$\mathbb{E}_{\theta}[\hat{\theta}(X)] = \mathbb{E}_{\theta}\left[\mathbb{E}_{\theta}\left[\frac{Y}{N}\middle|N\right]\right] = \mathbb{E}_{\theta}\left[\frac{N\theta}{N}\right] = \theta$$

and

$$\text{Var}_{\theta}[\hat{\theta}(X)] = \mathbb{E}_{\theta}[\text{Var}(\frac{Y}{N}|N)] + \text{Var}_{\theta}[\mathbb{E}(\frac{Y}{N}|N)] = \mathbb{E}_{\theta}\left[\frac{N\theta(1-\theta)}{N^2}\right] + \text{Var}_{\theta}[\theta] = \theta(1-\theta)\mathbb{E}\left[\frac{1}{N}\right].$$

Problem 2

Let $X \sim f_\theta(x)$, where $\theta \in \Theta$ is the unknown parameter. For any $\theta, \theta' \in \Theta$, let

$$\psi_{\theta',\theta}(x) := \frac{f_{\theta'}(x)}{f_\theta(x)} - 1.$$

i) Show that for any scalar statistic $T(x)$,

$$\mathbb{E}_\theta[\psi_{\theta',\theta}(X)T(X)] = \mathbb{E}_{\theta'}[T(X)] - \mathbb{E}_\theta[T(X)], \quad \forall \theta, \theta' \in \Theta$$

In particular, we have

$$\mathbb{E}_\theta[\psi_{\theta',\theta}] = 0, \quad \forall \theta, \theta' \in \Theta$$

ii) Use the Cauchy-Schwarz inequality to show that

$$\text{Var}_\theta[\hat{g}(X)] \geq \sup_{\theta' \in \Theta} \frac{(\phi(\theta') - \phi(\theta))^2}{\mathbb{E}_\theta[\psi_{\theta',\theta}^2(X)]}, \quad \forall \theta \in \Theta$$

for any estimator $\hat{g}(X)$ of $g(\theta)$, where

$$\phi(\theta) := \mathbb{E}_\theta[\hat{g}(X)].$$

Solution

Part (i): Integrating

$$\begin{aligned} \mathbb{E}_\theta[\psi_{\theta',\theta}(X)T(X)] &= \int \left(\frac{f_{\theta'}(x)}{f_\theta(x)} - 1 \right) T(x) f_\theta(x) dx \\ &= \int T(x) f_{\theta'}(x) dx - \int T(x) f_\theta(x) dx \\ &= \mathbb{E}_{\theta'}[T(X)] - \mathbb{E}_\theta[T(X)] \end{aligned}$$

Problem 3

Let $X \sim p_\theta(x)$ and $T = T(X)$ be a statistic for θ . Let $s_\theta(x)$ and $I_X(\theta)$ be the score function and Fisher information of θ with respect to the observation X , and let $\tilde{s}_\theta(t)$ and $I_T(\theta)$ be the score function and Fisher information of θ with respect to the statistic T .

i) Show that

$$\tilde{s}_\theta = \mathbb{E}_\theta[s_\theta(X)|T], \quad \forall \theta.$$

ii) Show that

$$I_T(\theta) \preceq I_X(\theta), \quad \forall \theta,$$

where the equalities hold when T is sufficient statistic of θ . Here, $A \preceq B$ means that $B - A$ is positive definite.

Solution

Part (i): The Fisher score

$$\tilde{s}_\theta = \int \frac{\partial}{\partial \theta} \log p_\theta(t) p_\theta(t) dt = \int \frac{\partial}{\partial \theta} p_\theta(t) dt = \mathbb{E}_\theta[s_\theta(X)|T], \quad \forall \theta.$$

Part (ii): By total variance,

$$I_X(\theta) = \mathbb{E}_\theta[\text{Var}(s_\theta(X)|T)] + \text{Var}_\theta[\mathbb{E}(s_\theta(X)|T)] = \mathbb{E}_\theta[\text{Var}(s_\theta(X)|T)] + I_T(\theta) \succeq I_T(\theta), \quad \forall \theta,$$

therefore

$$I_T(\theta) \preceq I_X(\theta), \quad \forall \theta$$

Problem 4

Let $X \sim f_\theta(x) = \frac{1}{\theta} g\left(\frac{x}{\theta}\right)$, where $\theta > 0$ is the unknown parameter and g is a fixed density function.

i) Show that

$$I(\theta) = \frac{1}{\theta^2} \mathbb{E}_g \left[\left(\frac{Y g'(Y)}{g(Y)} + 1 \right)^2 \right]$$

where the expectation is over $Y \sim g(y)$.

ii) Let $\eta = \log \theta$. Compute $I(\eta)$.

iii) Compute $I(\theta)$ when g is *Cauchy*(0,1).

Solution

Part (i): Here

$$\log f_\theta(x) = \log 1 - \log \theta + \log g\left(\frac{x}{\theta}\right)$$

Then differentiating

$$\frac{\partial}{\partial \theta} \log f_\theta(x) = -\frac{1}{\theta} + \frac{g'(\frac{x}{\theta})}{g(\frac{x}{\theta})} \cdot \left(-\frac{x}{\theta^2}\right) = \frac{1}{\theta} \left(\frac{Y g'(Y)}{g(Y)} + 1 \right)$$

By the Fisher information theorem,

$$I(\theta) = \frac{1}{\theta^2} \mathbb{E}_g \left[\left(\frac{Y g'(Y)}{g(Y)} + 1 \right)^2 \right]$$

Part (ii): Here

$$I(\eta) = -\mathbb{E}_\eta \left[\nabla_\eta^2 \log \tilde{f}_\eta(x) \right] = \frac{\theta^2}{\theta^2} \mathbb{E}_g \left[\left(\frac{Y g'(Y)}{g(Y)} + 1 \right)^2 \right] = \mathbb{E}_g \left[\left(\frac{Y g'(Y)}{g(Y)} + 1 \right)^2 \right]$$

Part (iii): Here

$$g(y) = \frac{1}{\pi(1+y^2)}, \quad g'(y) = -\frac{2y}{\pi(1+y^2)^2}$$

which gives

$$I(\theta) = \frac{1}{\theta^2} \mathbb{E}_g \left[\left(-\frac{y^2 - 1}{y^2 + 1} \right)^2 \right] = \frac{1}{\theta^2} \mathbb{E}_g \left[\left(\frac{y^2 - 1}{y^2 + 1} \right)^2 \right] = \frac{1}{\theta^2}$$